

REDCap

Advanced REDCap



- You're a Data Manager
  - Wrote DMP
  - Identified tools and methods
  - Developing data collection tools
    - Single tool/CRF/Instrument
  - Longitudinal studies?

Advanced REDCap – Ask students

- Longitudinal Projects
  - Multiple study sites
  - Multiple practitioners
  - Has multiple visits
  - Has multiple study/treatment arms/groups
  - Has multiple different tools/instrument/CRF
  - Study participants can be seen more than once for a particular visit, etc

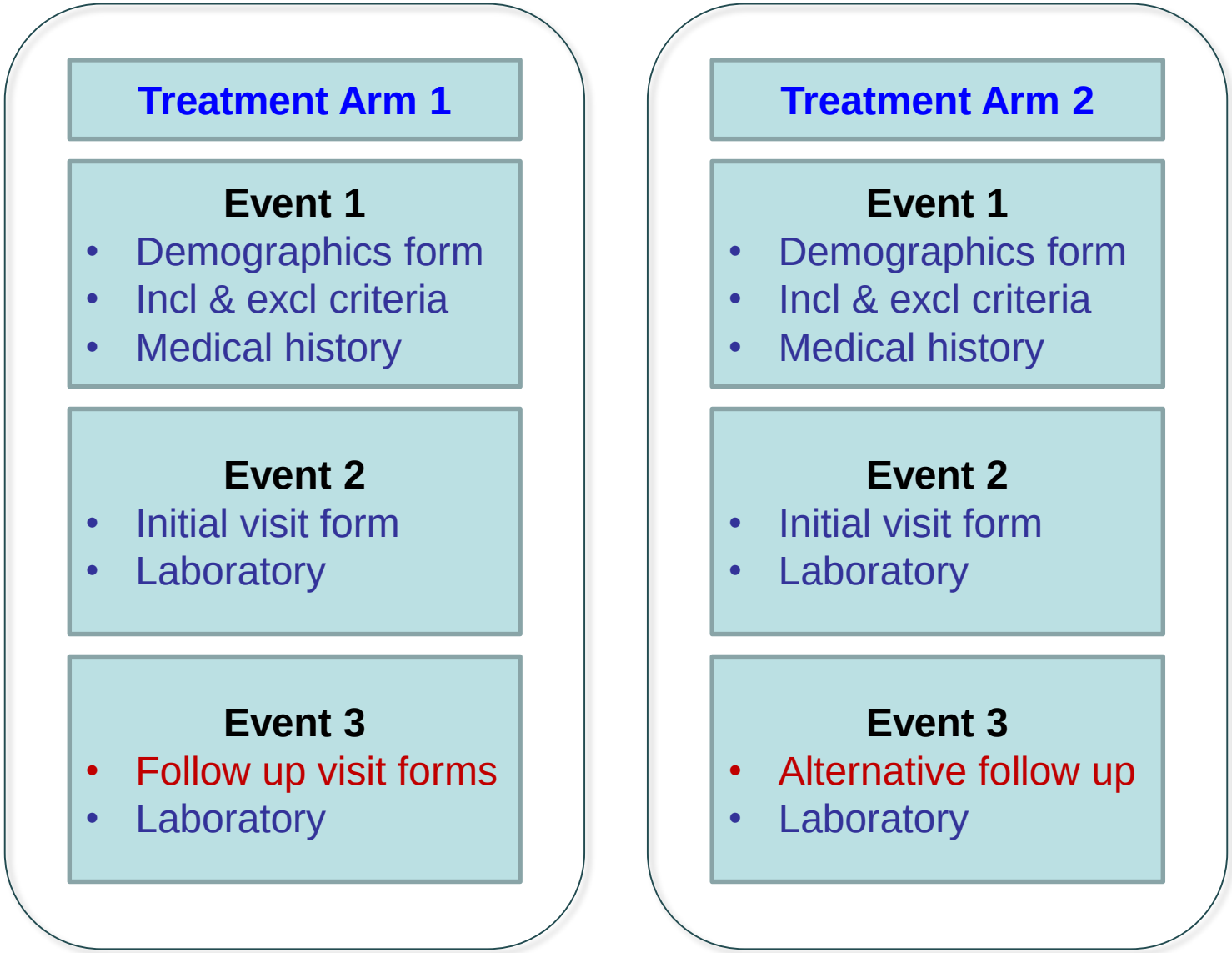
Longitudinal projects

- Purpose
  - Clinical trials
  - Multi-site studies(each site as a different arm)
  - Long term studies
  - Repeating Surveys (e.g every 3 or so months)

- Pros
  - Simple
  - Recycling (build form once and just re-use)
  - Scheduling feature
  - Flexibility (add more events)

- Cons
  - Complex
  - Large record status dashboard

REDCap structure

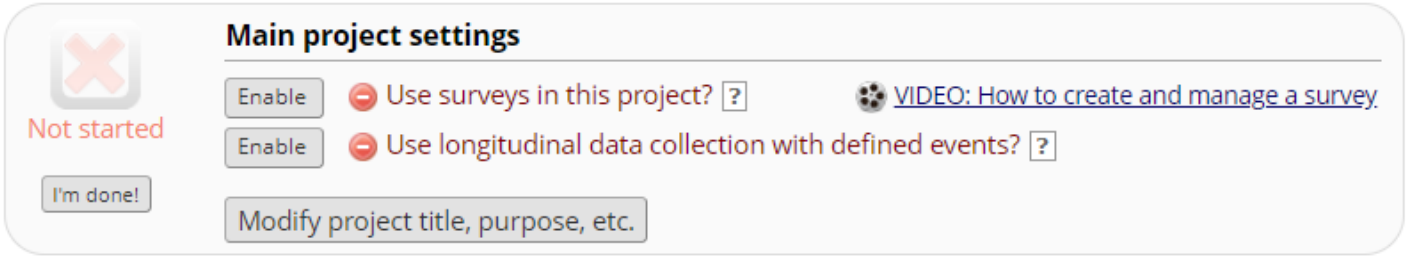


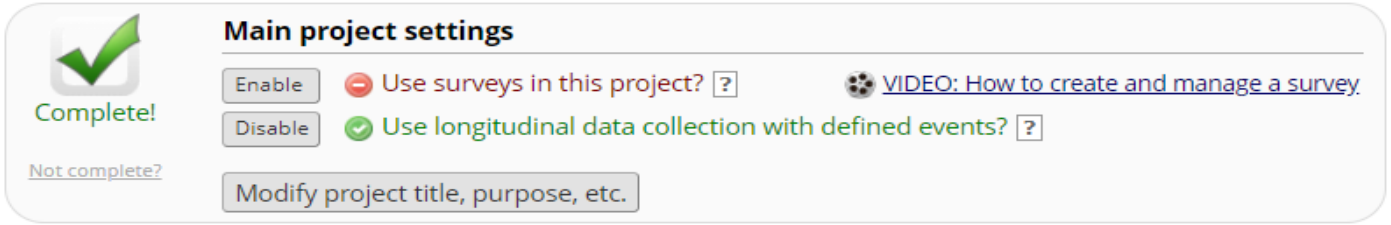
Steps for longitudinal utility

- Design all instruments
- Go to settings
- Enable longitudinal setup
- Define events and arms
- Designate instruments to events
- Test
- Convert to production mode
- Data collection
- Made changes on the instruments

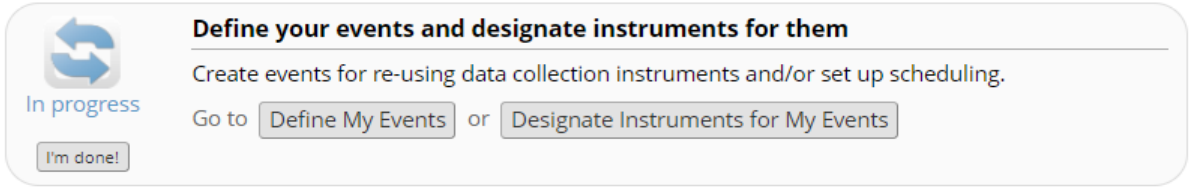
Enable longitudinal

- Click enable longitudinal data collection





- New section pop-up



Arms and events

- Arms
  - Arm name
  - Arm number – identifier
  - Can differ
    - Events
    - No. of forms

- Events
  - Unique event name: (auto-generated) i.e.
    - eventname_arm_1 e.g.
      - baseline_arm_1
  - Arm number – identifier
  - Days offset - offset Range (useful if scheduling feature enabled)
  - Each event can contain multiple forms
  - Basis for generating schedules

Note: Use DAGS or create separate projects for each site.

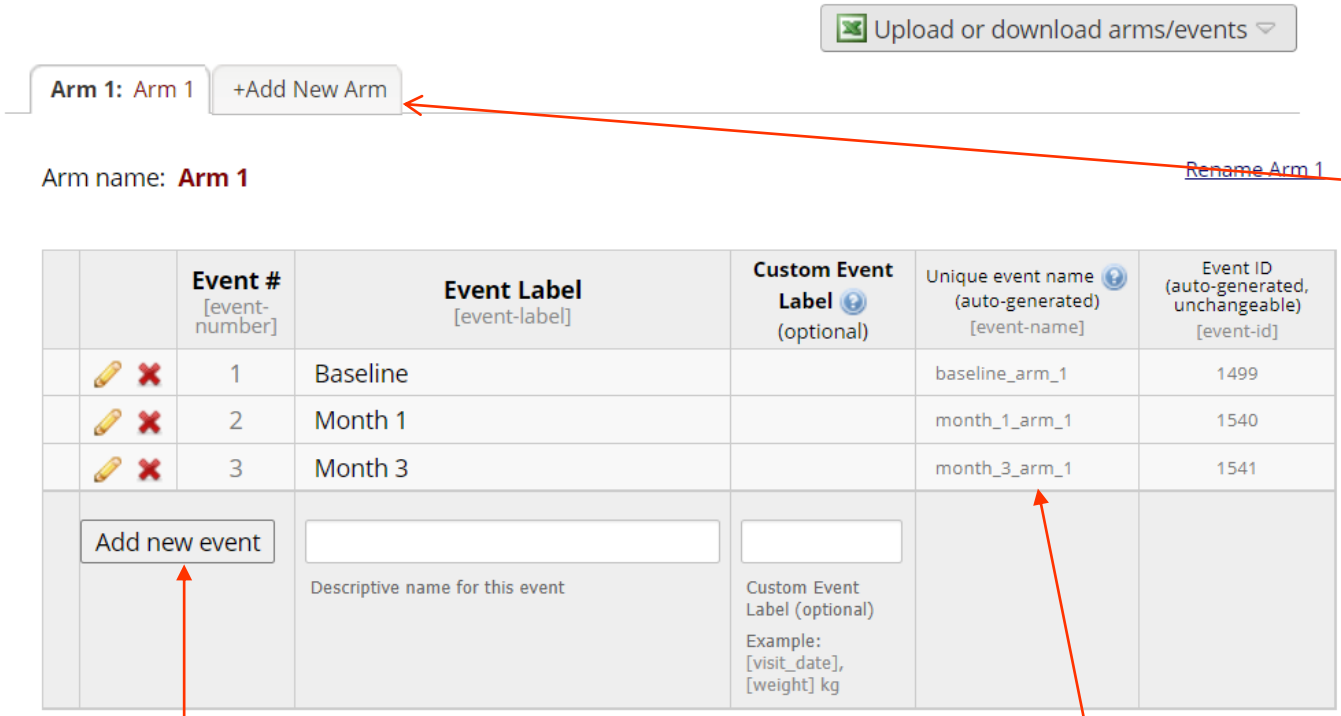
Data Access Groups (DAGs)

- Multi-centre studies
  - One project, one group per site
  - Users in a DAG see only their own site's records
  - Users without a DAG (data managers) see all records
- Advantages
  - Same instruments and codes at every site
  - One export for the pooled analysis
- Example: PISTIC 2026
  - Dar es Salaam, Mwanza, Mbeya: three DAGs

Setting up DAGs

- Applications → Data Access Groups
  - Create one DAG per site
- User Rights
  - Assign each site user to a DAG
  - Keep data managers without a DAG
- Record auto-numbering
  - New record IDs get the DAG prefix (e.g. 2-1)
- Data import and export
  - Column redcap_data_access_group (unique group name)
  - Reports can be filtered by DAG

Defining events



If two or more arms click add Arm

Add event name/description and click Add new event

Unique event name is autogenerated

- Event name
- Arm

Designate instruments to events

Upload or download instrument mappings

Begin Editing | Save | Select All | Deselect All

Data Collection Instrument	Baseline (1)	Month 1 (2)	Month 3 (3)	Month 6 (4)
Locator	☐	☐	☐	☐
Patient Info	☐	☐	☐	☐
Sexual history	☐	☐	☐	☐
Pregnancy form	☐	☐	☐	☐
Pregnancy outcome	☐	☐	☐	☐
Concomitant med	☐	☐	☐	☐
Vaccination 1	☐	☐	☐	☐
Vaccination 2	☐	☐	☐	☐

Data Collection Instrument	Baseline (1)	Month 1 (2)	Month 3 (3)	Month 6 (4)
Locator	☑	☐	☐	☐
Patient Info	☑	☐	☐	☐
Sexual history	☑	☑	☑	☑
Pregnancy form	☑	☑	☑	☑
Pregnancy outcome	☑	☑	☑	☑
Concomitant med	☑	☑	☑	☑
Vaccination 1	☐	☐	☑	☐
Vaccination 2	☐	☐	☐	☑

Data Collection Instrument	Baseline (1)	Month 1 (2)	Month 3 (3)	Month 6 (4)
Locator	✔			
Patient Info	✔			
Sexual history	✔	✔	✔	✔
Pregnancy form	✔	✔	✔	✔
Pregnancy outcome	✔	✔	✔	✔
Concomitant med	✔	✔	✔	✔
Vaccination 1			✔	
Vaccination 2				✔

Final display

Data entry in presence of arms

Add / Edit Records

You may view an existing record/response by selecting it from the drop-down lists below. To create a new record/response, type a new value in the text box below and hit Tab or Enter. To quickly find a record without using the drop-downs, the text box will auto-populate with existing record names as you begin to type in it, allowing you to select it.

Total records: **1**		
Choose an existing Participant ID	Arm 1: Arm 1	-- select record --
Enter a new or existing Participant ID	Arm 1: Arm 1	

Data Search	
Choose a field to search (excludes multiple choice fields)	All fields
Search query Begin typing to search the project data, then click an item in the list to navigate to that record.	

Record "MEPI001" is a new Participant ID. To create the record and begin entering data for it, click any gray status icon below.

The grid below displays the form-by-form progress of data entered for the currently selected record. You may click on the colored status icons to access that form/event. If you wish, you may modify the events below by navigating to the Define My Events page.

Legend for status icons:
- Incomplete
- Incomplete (no data saved) ?
- Unverified
- Complete

NEW Participant ID **MEPI001**

Data Collection Instrument	Baseline	Month 1	Month 3	Month 6
Locator	●			
Patient Info	●			
Sexual history	●	●	●	●
Pregnancy form	●	●	●	●
Pregnancy outcome	●	●	●	●
Concomitant med	●	●	●	●
Vaccination 1			●	
Vaccination 2				●

Data entry - example

Treatment A

- Enrollment
- Month 1 – 12 (Every 3 months)
- End of study

Treatment B

- Enrollment
- Month 1 - 12, Every 6 months
- End of study

Record Status Dashboard

Participant ID **MEPI001**
Arm 1: Treatment A

Data Collection Instrument	Baseline	Month 1	Month 3	Month 6	Month 9	Month 12
Demographics						
LQ						
Lab Specimen Tracker						
Drug Dispensing Log						
Delete all data on event:						

Participant ID **MEPI004**
Arm 2: Treatment B

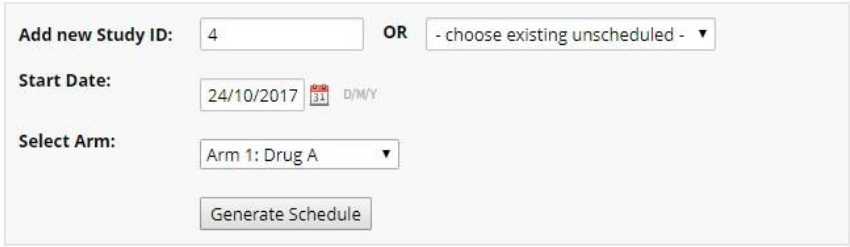
Data Collection Instrument	Baseline	Month 1	Month 6	Month 12
Demographics				
LQ				
Lab Specimen Tracker				
Drug Dispensing Log				
Delete all data on event:				

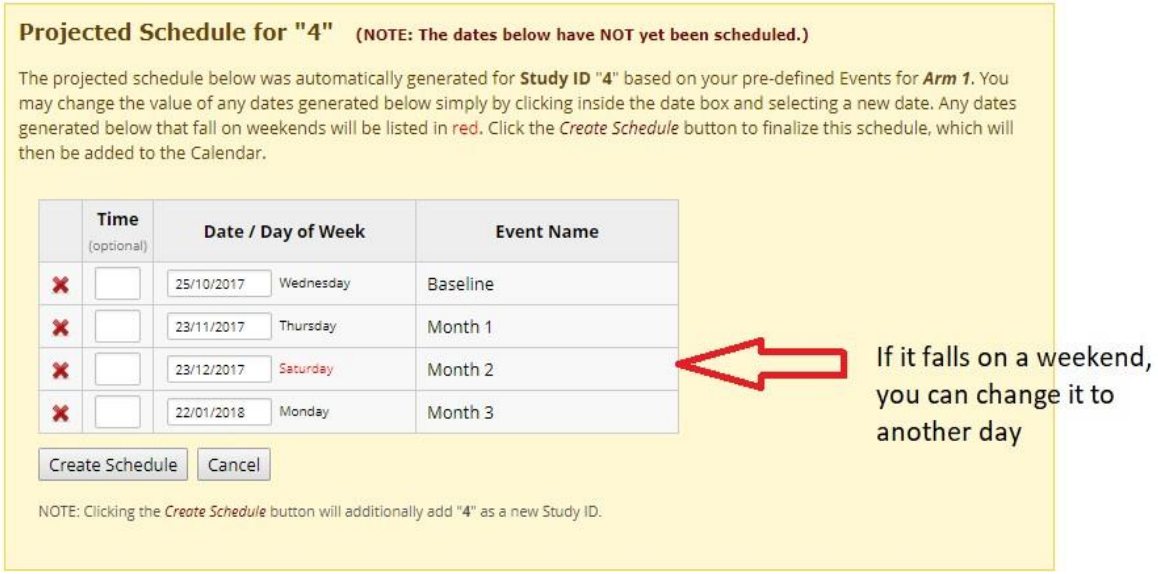
Calendar

- Automated generated
  - Per participant
  - Based on arms and events
  - Advantage: Linked to outside calendar

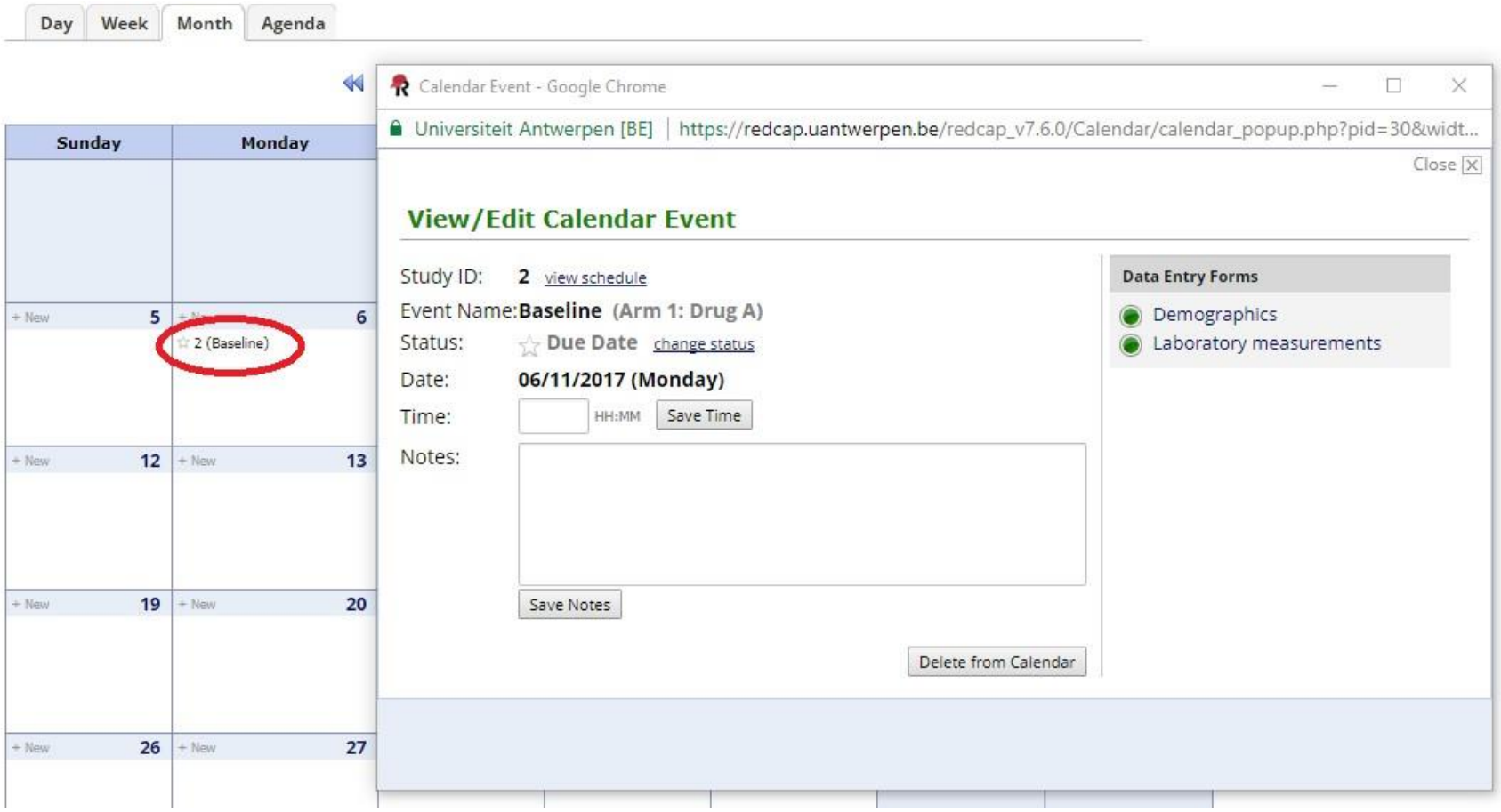
  (e.g. outlook, google calendars)

- Multiple views i.e. by date, month or year
- Event status
  - e.g. scheduled, cancelled visits etc.
- Customize user rights





Calendar – event status



Branching logic

- Classical
  - Define the variable
  - Put in an operator
  - Specify you comparison value

    [age_of_child] >= ‘18’
- Longitudinal
  - Define event name
  - Define the variable
  - Put in an operator
  - Specify you comparison value

    [baseline_arm_1][age_of_child] >= ‘18’

Field embedding

- What
  - Place a field inside the label of another field
  - Write the variable name in curly brackets: {weight_kg}
  - Only on the same instrument (same survey page)
- Example: a table in a descriptive field
  - <table><tr><td>Weight {weight_kg}</td><td>Height {height_cm}</td></tr></table>
- Use for
  - Grids and tables (signs, test results), compact forms

Piping

- What
  - Insert a value already collected into text: [variable]
- Examples
  - Of these [sex_times] times, how many times was a condom used?
  - [sex] shows the label, [sex:value] the code
  - [baseline_arm_1][age]: value from another event
- Where
  - Field labels and notes, section headers, choice labels
  - Survey instructions, invitations, @DEFAULT

Action tags

- Written in the Field Annotation
  - @TODAY, @NOW: today's date or time
  - @HIDDEN, @HIDDEN-SURVEY, @READONLY
  - @CHARLIMIT=n, @WORDLIMIT=n
  - @CALCTEXT(…), @CALCDATE(…)
  - @HIDECHOICE='3', @NONEOFTHEABOVE
  - @LATITUDE, @LONGITUDE, @DEFAULT='…'

Smart variables and special functions

- Smart variables
  - [record-name], [event-name], [arm-number]
  - [first-event-name], [record-dag-name], [user-name]
- Special functions
  - rounddown(datediff([dob], 'today', 'y'))
  - if([weight] > 100, 1, 0)
  - round(), sum(), mean(), isblankormissingcode()

Surveys

- No login needed
- Multiple/single surveys
- Useful
  - Recruitment
  - Participant data entry
- How to send surveys (ways)
  - Public links
  - Notifications
  - Automatic invites

Public links vs Personal Invitation

- Public links
  - Ideal for recruitment
  - Initially anonymous(unless you ask for names)
  - Filled multiple times
  - Auto-generated by REDCAP (short link available)
  - Only available for the first survey of each
  - arm(Must contain unique_id)

- Personal invites
  - Define event name
  - Email required
  - Identifiable
  - Traceable completion
  - Customizable invitations

Manage survey participants

Survey Distribution Tools

Public Survey Link | Participant List | Survey Invitation Log

Using a public survey link is the simplest and fastest way to collect responses for your survey. You may obtain the survey link below to email it to your participants. Responses will be collected anonymously (unless the survey contains questions asking for identifying data from the participant). **NOTE:** Since this method uses a single survey link for all participants, it allows for the possibility of participants taking the survey multiple times, which may be necessary in some cases.

To obtain the survey link, copy the URL below and paste it into the body of an email message in your own email client. Your email recipient(s) can then click the link to begin taking your survey.

Public Survey URL: https://redcap.uantwerpen.be/surveys/?s=88EEXCX4DA3TWTRX

Link Actions

- Open public survey
- Open public survey + Log out
- Send me URL via email
- Survey Access Code or QR Code

Link Customizations

- Get Short Survey Link
- Create Custom Survey Link
- Get Embed Code

Participant List belonging to [Initial survey] "Demographics" - Baseline (Arm 1: Drug A)

Remove all participants

Displaying 1 - 2 of 2 | Add participants | Compose Survey Invitations | Export list

Email	Participant Identifier (Enable)	Responded?	Invitation Scheduled?	Invitation Sent?	Link	Survey Access Code and QR Code
[No email listed]	Disabled		-			
[No email listed]	Disabled		-			

Survey Invitation Log

(in ascending order by time sent)

View past invitations | View future invitations

Displaying 0 of 0

Begin time: 31/10/2017 19:02 End time: (D/M/Y H:M)

Display All invitation types and All response statuses

Display All surveys

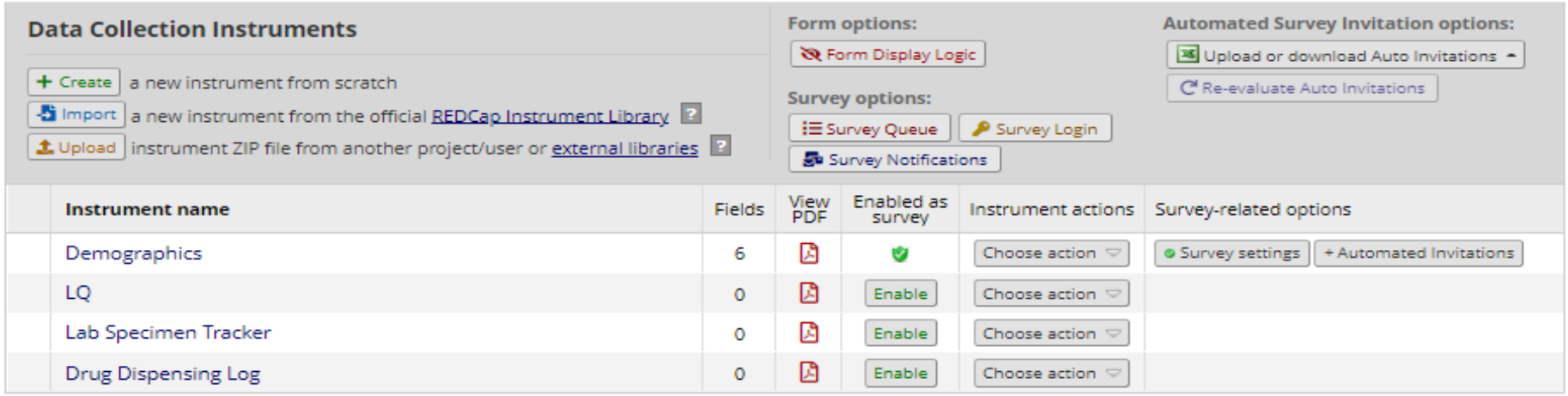
Display invitation reminders?

Apply filters | Reset | Download log (as seen below)

Invitation send time	View Invite	Participant Email	Participant Identifier	Survey	Survey Link	Responded?	Errors (if any)
No invitations to list							

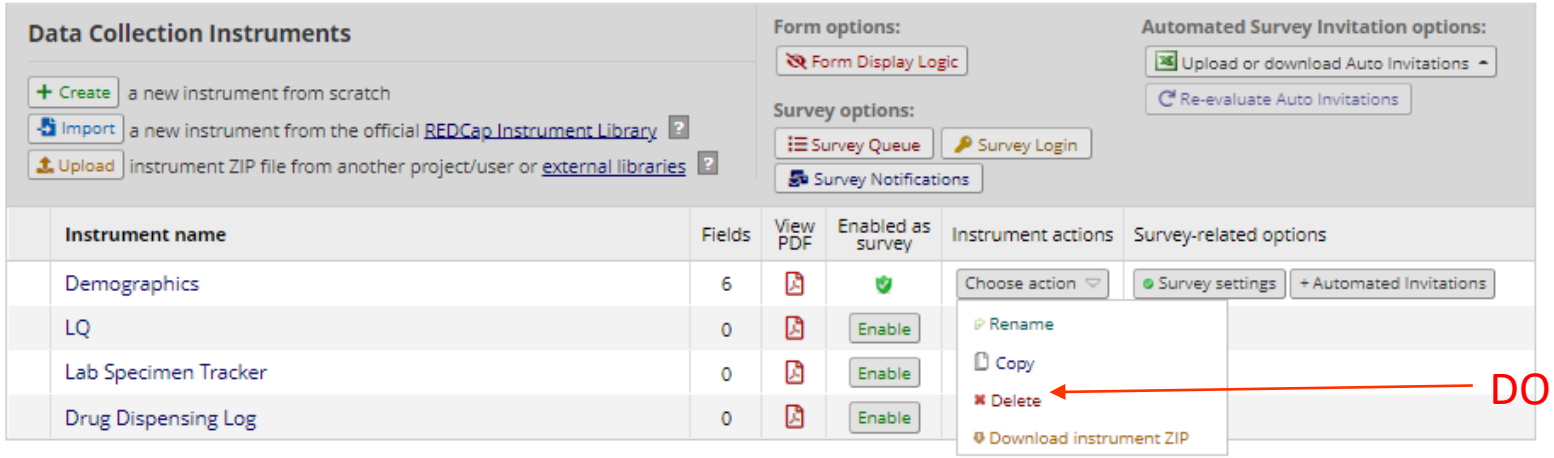
Setting up the surveys

- Enable in main project settings
- Create forms
- Enable forms as surveys
- Set settings
- Invite participants



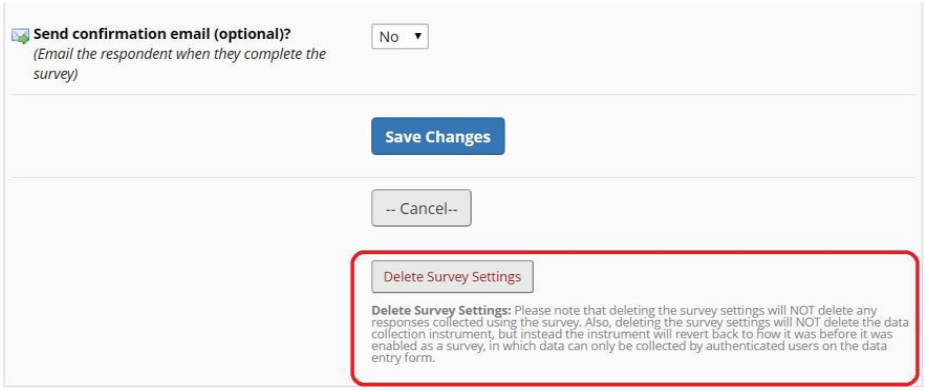
Disabling surveys

- NOT in the form actions



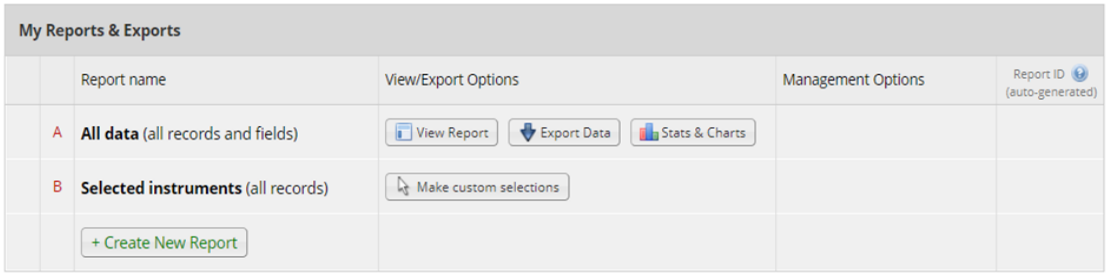
DONOT USE THIS DELETE OPTION

- Use survey settings
- Click ‘delete survey settings



Statistics, charts and reports

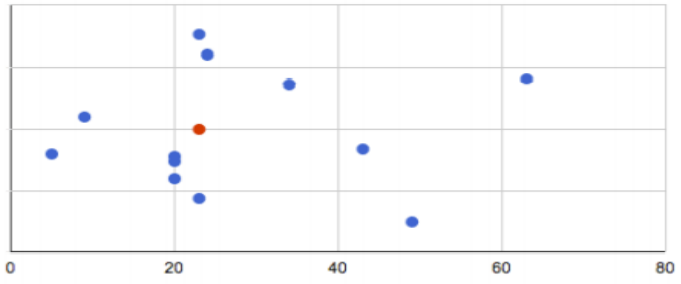
- Data can be viewed for quick analysis inside of REDCap using the “Stats and charts” feature
- Option to choose report for all data or specific record or instrument.

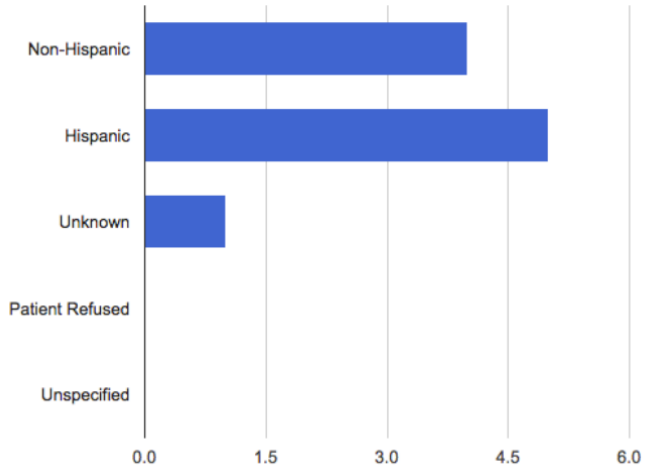


Age Refresh Plot

Total Count (N)	Missing	Unique	Min	Max	Mean	StDev	Sum	Percentile						
								0.05	0.10	0.25	0.50 Median	0.75	0.90	0.95
12	10 (45.5%)	9	5.00	63.00	27.75	16.72	333.00	7.20	10.10	20.00	23.00	36.25	48.40	55.30

Lowest values: 5, 9, 20, 20, 20
Highest values: 24, 34, 43, 49, 63





Practicals

- Login to REDCap: https://redcap.uantwerpen.be/
- Use the questionnaire sent on blackboard
- Develop database for this questionnaire
- Use Online designer
- Import using data dictionary
- Move your project to production mode
- Check user rights
- Change your questionnaire to surveys
- Use mobile app
- Try create a longitudinal project by just defining the instruments, arm and event
- Use scheduler

Extra materials

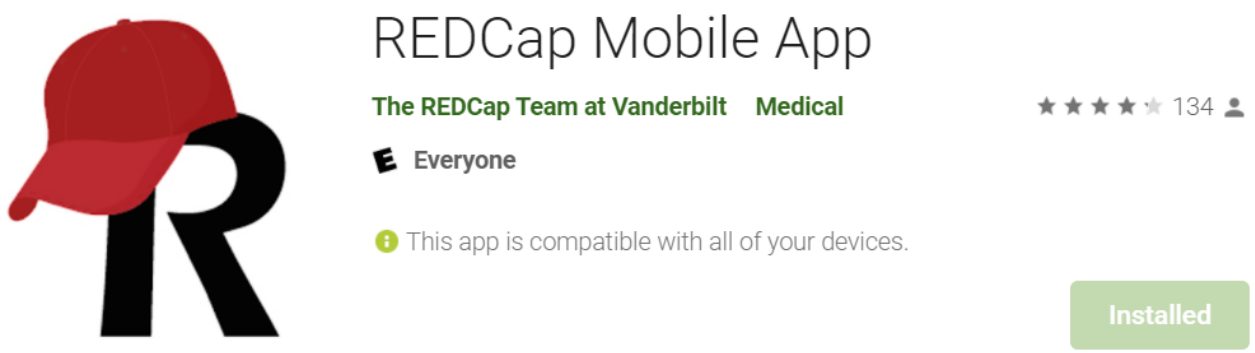
- REDCap Training videos: https://projectredcap.org/resources/videos/
- REDCap videos: https://cctsi.cuanschutz.edu/resources/informatics/redcap-resources

Package ‘REDCapR’

- Pulling data directly from REDCap using R

Mobile App

- Available on Android and iOS systems
- Used for offline data collection
- Connected to the server



Homework: PISTIC 2026

- Build the PISTIC study in REDCap (groups)
  - Protocol and CRFs: REDCap Protocol and CRFs – Assignment
  - Instructions: REDCap HW Instructions 2026
- New this year
  - Multi-centre: Dar es Salaam, Mwanza, Mbeya
  - Data Access Groups: one per site
- Deadline
  - **13 December 2026, 11:00**

End

Setting up class demonstration project (20Nov2026)

- Login in to your REDCap account
- Go to <New Project>
- Complete the <Create a new REDCap Project>
  - Project Title <longitudinal studies demo – s2#######>; where s2#######> is your student number
  - Project's purpose: <Practice/ Just for fun>
  - Project creation option: <Upload a REDCap project XML file>
  - Choose the XML file from the folder you unzipped the XML file
- Request Project

➢ Note: We'll use this project to demonstrate how longitudinal studies is set and also explore other advanced features in REDCap.

➢ Make sure you create this project before the class.